SYED UMAR

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EDUCATION

Presidency University, Bangalore, Karnataka, India — Aug 2024 – May 2028

Bachelor of Technology, Computer Science and Engineering | CGPA: 9.33 / 10.0

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems, Computer Networks, Discrete Mathematics
- Actively pursuing Harvard University's CS50x (completed), CS50 Web Programming, and CS50 AI certifications in parallel with undergraduate studies

HONORS & AWARDS

1st Place — Industry-Hosted Hardware Hackathon 2026

- Designed and built *SewerOne*, a semi-autonomous sewage-cleaning robot, winning among all competing university teams; judged on engineering rigor, real-world applicability, and hardware-software integration.

Top 9 of 207 Teams — Smart India Hackathon Internal Selection 2025

- Advanced *SoulSpace*, an AI-assisted mental health companion, to the national shortlist; evaluated by industry experts on clinical relevance, AI architecture, and scalability.

Harvard CS50x Certificate — with Distinction 2025

- Completed Harvard's flagship computer science program covering C, Python, SQL, web fundamentals, and algorithmic thinking; currently pursuing CS50W and CS50 AI.

PROJECTS

SewerOne — Autonomous Sewage-Cleaning Robot 2026

ROS2 Humble · Nav2 · SLAM · Gazebo · RViz2 · LiDAR · Python · C++

- Engineered full autonomous navigation stack on ROS2 using Nav2, SLAM Toolbox, and LiDAR; robot localises itself in unknown sewage environments and plans collision-free paths in real time.
- Simulated the complete robot environment in Gazebo and visualised sensor data and pose estimates in RViz2 before deploying to hardware, reducing physical test cycles by ~60%.
- Won 1st place at an industry-hosted hackathon; solution addresses a critical sanitation problem affecting millions, replacing dangerous manual labour.

SoulSpace — AI Mental Health Companion Platform 2025

Python · Django · NLP · PHQ-9 / GAD-7 · REST API · PostgreSQL

- Built a clinically-grounded mental health monitoring platform integrating PHQ-9 and GAD-7 standardised assessment frameworks to quantify depression and anxiety severity scores.
- Implemented NLP-driven conversational analysis to detect emotional patterns and surface personalised coping recommendations; designed for scale to serve concurrent users.
- Ranked Top 9 of 207 teams in Smart India Hackathon internal selection, recognised for clinical accuracy, responsible AI design, and system architecture.

TreeLan — AI-Powered Freelance Economy Platform 2025

React.js · Django · REST API · AI Assessment Engine · PostgreSQL

- Designed a skill-first hiring marketplace where candidates are evaluated through AI-generated adaptive assessments and verified portfolio demonstrations rather than résumé proxies.
- Built end-to-end: React.js frontend with responsive UI, Django REST backend, and an AI assessment engine that dynamically adjusts question difficulty based on prior responses.
- Architected database schema and API layer supporting multi-role users (freelancers, clients, admins) with role-based access control and JWT authentication.

CoinologicServices — Production Business Web Platform 2026

Django · Flask · Google OAuth 2.0 · SEO · PostgreSQL · Python

- Deployed a full-stack business website handling real user traffic; integrated Google OAuth 2.0 for zero-friction authentication with secure session management and CSRF protection.
- Implemented on-page SEO optimisation (structured data, meta tags, sitemap) resulting in measurable improvements in organic search visibility.
- Built responsive, accessible UI ensuring cross-device compatibility; maintained 99%+ uptime on production deployment.

CS50W Full-Stack Application Suite 2025–2026

Django · Python · JavaScript · SQLite · HTML/CSS

- Developed a suite of five production-grade Django applications: Google-style search engine, collaborative wiki/encyclopedia, eBay-style auction platform, and a full email client — each with complete backend, database models, and frontend.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript, HTML, CSS

Frameworks: Django, Flask, React.js, ROS2 (Humble), Nav2

Robotics & AI: SLAM, LiDAR Integration, Gazebo, RViz2, Autonomous Navigation, NLP

Databases: PostgreSQL, SQLite, MySQL

Tools & Infra: Git, GitHub, Linux/Ubuntu, REST APIs, OAuth 2.0, JWT, SEO

Core Areas: Backend Engineering, Full-Stack Development, Autonomous Systems, AI-Assisted Applications